

G11 ADV Term 3
Academic Year 2025–2026

G11 ADV – EOT Revision

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Name:

Find the component form of $\overrightarrow{AB}$ with initial point $A(-4, 2)$ and terminal point $B(3, -5)$.

A	$\langle -1, -3 \rangle$
B	$\langle 7, -7 \rangle$
C	$\langle -7, 7 \rangle$
D	$\langle 1, 3 \rangle$

Find the component form of $\overrightarrow{AB}$ with the given initial and terminal points.

1A. $A(-2, -7), B(6, 1)$

A	$\langle 8, 8 \rangle$
B	$\langle -9, -11 \rangle$
C	$\langle 8, -6 \rangle$
D	$\langle 9, -11 \rangle$

Find the component form of $\overrightarrow{AB}$ with the given initial and terminal points.

1B. $A(0, 8), B(-9, -3)$

A	$\langle 8, 8 \rangle$
B	$\langle -9, -11 \rangle$
C	$\langle 8, -6 \rangle$
D	$\langle 9, -11 \rangle$

Find the component form of the vector v with magnitude 10 and direction angle 120° .

A	$\langle -5, -5\sqrt{3} \rangle$
B	$\langle -5, 5\sqrt{3} \rangle$
C	$\langle 4\sqrt{2}, 4\sqrt{2} \rangle$
D	$\langle -4\sqrt{2}, 4\sqrt{2} \rangle$

Find the component form of v with the given magnitude and direction angle.

6A. $|v| = 8, \theta = 45^\circ$

A	$\langle -5, -5\sqrt{3} \rangle$
B	$\langle -5, 5\sqrt{3} \rangle$
C	$\langle 4\sqrt{2}, 4\sqrt{2} \rangle$
D	$\langle -4\sqrt{2}, 4\sqrt{2} \rangle$

Find the component form of v with the given magnitude and direction angle.

6B. $|v| = 24, \theta = 210^\circ$

A	$\langle -5, -5\sqrt{3} \rangle$
B	$\langle -5, 5\sqrt{3} \rangle$
C	$\langle -12\sqrt{3}, -12 \rangle$
D	$\langle -12\sqrt{3}, 12 \rangle$

Find the angle θ between vectors u and v to the nearest tenth of a degree.

a. $u = \langle 6, 2 \rangle$ and $v = \langle -4, 3 \rangle$

A	$\theta = 63.4$
B	$\theta = 121.3$
C	$\theta = 45$
D	$\theta = 161.6$

b. $u = \langle 3, 1 \rangle$ and $v = \langle 3, -3 \rangle$

A	$\theta = 63.4$
B	$\theta = 121.3$
C	$\theta = 100.6$
D	$\theta = 161.6$

3A. $u = \langle -5, -2 \rangle$ and $v = \langle 4, 4 \rangle$

A	$\theta = 63.4$
B	$\theta = 121.3$
C	$\theta = 100.6$
D	$\theta = 161.6$

3B. $u = \langle 9, 5 \rangle$ and $v = \langle -6, 7 \rangle$

A	$\theta = 63.4$
B	$\theta = 121.3$
C	$\theta = 100.6$
D	$\theta = 161.6$

Find the dot product of u and v . Then determine if u and v are orthogonal.

a. $u = \langle -7, 3, -3 \rangle, v = \langle 5, 17, 5 \rangle$

A	0 ; Orthogonal
B	2;not Orthogonal
C	-2;not Orthogonal
D	-6;not Orthogonal

b. $u = \langle 3, -3, 3 \rangle, v = \langle 4, 7, 3 \rangle$

A	0 ; Orthogonal
B	2;not Orthogonal
C	-2;not Orthogonal
D	-6;not Orthogonal

4	Find dot products of and angles between vectors in space.	Example 1 & (1A,1B)	446
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Find the dot product of $\mathbf{u}$ and $\mathbf{v}$. Then determine if $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

1A. $\mathbf{u} = \langle 3, -5, 4 \rangle, \mathbf{v} = \langle 5, 7, 5 \rangle$

1B. $\mathbf{u} = \langle 4, -2, -3 \rangle, \mathbf{v} = \langle 1, 3, -2 \rangle$

A	0 ; Orthogonal
B	2;not Orthogonal
C	-2;not Orthogonal
D	-6;not Orthogonal

A	0 ; Orthogonal
B	1;not Orthogonal
C	-1;not Orthogonal
D	-6;not Orthogonal

5	Find cross products of vectors in space, and use cross products to find area and volume.	Example 5 & (5)	449
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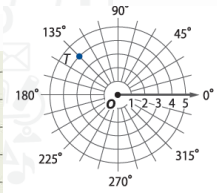
Find the area of the parallelogram with adjacent sides $\mathbf{u} = 2\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$ and $\mathbf{v} = \mathbf{i} - 5\mathbf{j} + 3\mathbf{k}$.

A	$\sqrt{545}=23.3$
B	$\sqrt{286}=16.9$
C	$\sqrt{214}=14.6$
D	$\sqrt{317} = 17.8$

5. Find the area of the parallelogram with adjacent sides $\mathbf{u} = -6\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{v} = 4\mathbf{i} + 3\mathbf{j} + \mathbf{k}$.

A	$\sqrt{545}=23.3$
B	$\sqrt{286}=16.9$
C	$\sqrt{214}=14.6$
D	$\sqrt{317} = 17.8$

Find four different pairs of polar coordinates that name point T if $-360^\circ \leq \theta \leq 360^\circ$.



A	$(4, 135^\circ), (4, -225^\circ), (-4, 315^\circ), (-4, -45^\circ)$
B	$(4, 135^\circ), (4, -135^\circ), (-4, 45^\circ), (-4, -45^\circ)$
C	$(4, 45^\circ), (4, -315^\circ), (-4, 225^\circ), (-4, -135^\circ)$
D	$(4, 135^\circ), (4, -225^\circ), (-4, 45^\circ), (-4, -315^\circ)$

Find three additional pairs of polar coordinates that name the given point if $-360^\circ \leq \theta \leq 360^\circ$ or $-2\pi \leq \theta \leq \pi$.

3A. $(5, 240^\circ)$

A	$(5, -120^\circ), (-5, 60^\circ), (-5, -300^\circ)$
B	$(5, -240^\circ), (-5, 60^\circ), (-5, -60^\circ)$
C	$(5, -120^\circ), (-5, 120^\circ), (-5, -240^\circ)$
D	$(5, 120^\circ), (-5, -60^\circ), (-5, -120^\circ)$

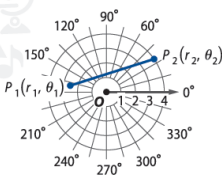
3B. $(2, \frac{\pi}{6})$

A	$(2, -\frac{5\pi}{6}), (-2, \frac{7\pi}{6}), (-2, -\frac{\pi}{6})$
B	$(2, -\frac{11\pi}{6}), (-2, \frac{7\pi}{6}), (-2, -\frac{5\pi}{6})$
C	$(2, -\frac{11\pi}{6}), (-2, \frac{5\pi}{6}), (-2, -\frac{7\pi}{6})$
D	$(2, -\frac{\pi}{6}), (-2, \frac{11\pi}{6}), (-2, -\frac{11\pi}{6})$

KeyConcept Polar Distance Formula

If $P_1(r_1, \theta_1)$ and $P_2(r_2, \theta_2)$ are two points in the polar plane, then the distance P_1P_2 is given by

$$\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_2 - \theta_1)}$$



Find the distance between each pair of points. (Example 5)

30. $(2, 30^\circ), (5, 120^\circ)$

A	10.7
B	0
C	8
D	5.68

31. $(3, \frac{\pi}{2}), (8, \frac{4\pi}{3})$

A	10.7
B	0
C	8
D	5.68

32. $(6, 45^\circ), (-3, 300^\circ)$

A	10.7
B	0
C	8
D	5.68

33. $(7, -\frac{\pi}{3}), (1, \frac{2\pi}{3})$

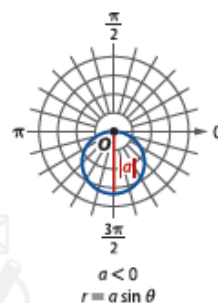
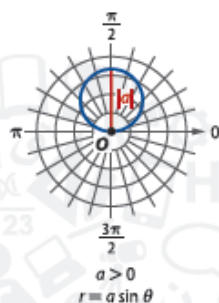
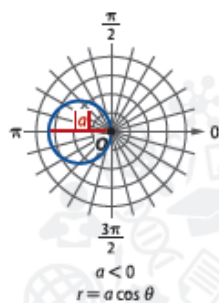
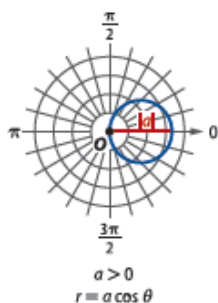
A	10.7
B	0
C	8
D	5.68

8	Identify and graph classical curves.	Concept Summary • Special Types of Polar Graphs	496
		(35-40)	498

ConceptSummary Special Types of Polar Graphs

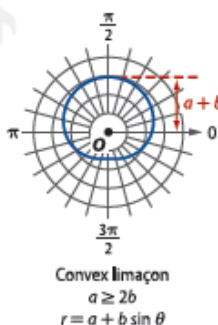
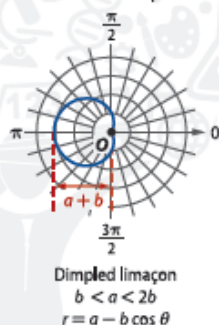
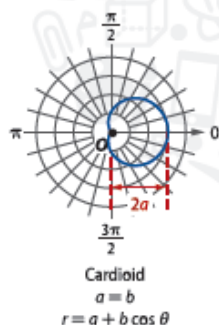
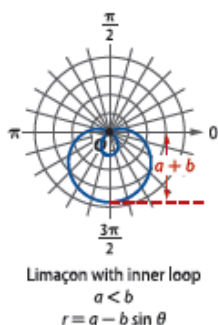
Circles

$$r = a \cos \theta \text{ or } r = a \sin \theta$$



Limaçons

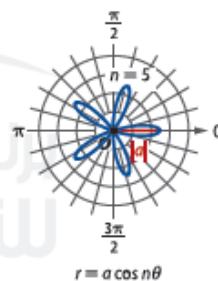
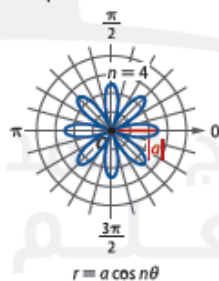
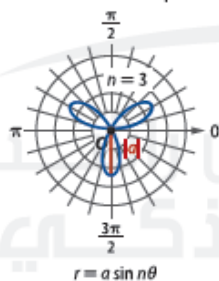
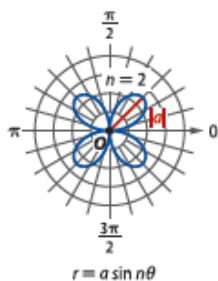
$$r = a \pm b \cos \theta \text{ or } r = a \pm b \sin \theta, \text{ where } a \text{ and } b \text{ are both positive}$$



Roses

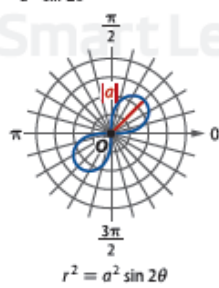
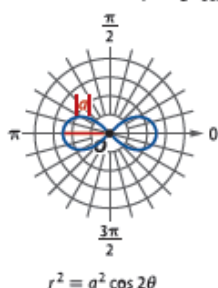
$$r = a \cos n\theta \text{ or } r = a \sin n\theta, \text{ where } n \geq 2 \text{ is an integer}$$

The rose has n petals if n is odd and $2n$ petals if n is even.



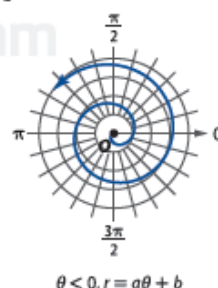
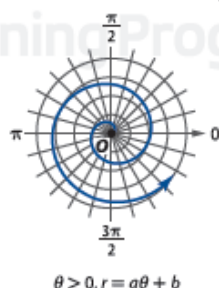
Lemniscates (LEM-nis-keyts)

$$r^2 = a^2 \cos 2\theta \text{ or } r^2 = a^2 \sin 2\theta$$



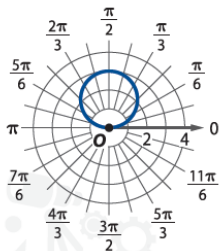
Spirals of Archimedes (ahr-kuh-MEE-deez)

$$r = a\theta + b$$



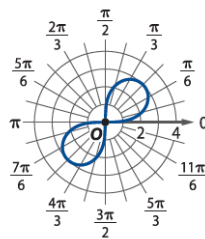
Write an equation for each graph.

35.



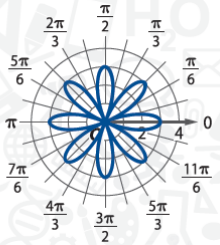
A	$r = 9\sin\theta$
B	$r = 3\sin\theta$
C	$r = 9\cos\theta$
D	$r = 3\cos\theta$

36.



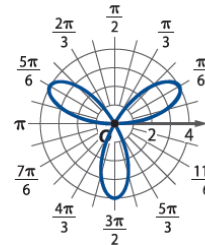
A	$r^2 = 9\sin 2\theta$
B	$r^2 = 3\sin 2\theta$
C	$r^2 = 9\sin 2$
D	$r^2 = 3\sin\theta$

37.



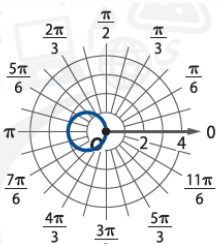
A	$r = 3\sin 4\theta$
B	$r = 3\cos 8\theta$
C	$r = 3\cos 4\theta$
D	$r = 4\cos 8\theta$

38.



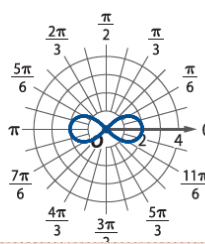
A	$r = 4\cos 3\theta$
B	$r = 4\cos 6\theta$
C	$r = 4\sin 3\theta$
D	$r = 4\sin 6\theta$

39.



A	$r = -2\sin\theta$
B	$r = 2\sin\theta$
C	$r = 2\cos\theta$
D	$r = -2\cos\theta$

40.



A	$r^2 = 4\sin 2\theta$
B	$r^2 = 4\sin\theta$
C	$r^2 = 4\cos 2\theta$
D	$r^2 = 4\cos\theta$

Find the rectangular coordinates for each point with the given polar coordinates.

a. $P\left(4, \frac{\pi}{6}\right)$

A	$(\sqrt{2}, -\sqrt{2})$
B	$(2\sqrt{3}, 2)$
C	$(3, -3\sqrt{3})$
D	$\left(-\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right)$

b. $Q(-2, 135^\circ)$

A	$(\sqrt{2}, -\sqrt{2})$
B	$(2\sqrt{3}, 2)$
C	$(3, -3\sqrt{3})$
D	$\left(-\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right)$

c. $V(3, -120^\circ)$

A	$(\sqrt{2}, -\sqrt{2})$
B	$(2\sqrt{3}, 2)$
C	$(3, -3\sqrt{3})$
D	$\left(-\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right)$

1A. $R(-6, -120^\circ)$

A	$(\sqrt{2}, -\sqrt{2})$
B	$(2\sqrt{3}, 2)$
C	$(3, -3\sqrt{3})$
D	$\left(-\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right)$

1B. $S\left(5, \frac{\pi}{3}\right)$

A	$\left(-\frac{3\sqrt{2}}{2}, -\frac{2\sqrt{3}}{2}\right)$
B	$(2\sqrt{3}, 2)$
C	$(3, -3\sqrt{3})$
D	$\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$

1C. $T(-3, 45^\circ)$

A	$\left(-\frac{3\sqrt{2}}{2}, -\frac{2\sqrt{3}}{2}\right)$
B	$(2\sqrt{3}, 2)$
C	$(3, -3\sqrt{3})$
D	$\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$

Identify the graph of each rectangular equation. Then write the equation in polar form. Support your answer by graphing the polar form of the equation.

a. $(x - 4)^2 + y^2 = 16$

A	Circle; $r = 8\cos\theta$
B	Circle; $r = 4\cos\theta$
C	Parabola; $r = \tan\theta \sec\theta$
D	Hyperbola; $r = \sec^2\theta$

b. $y = x^2$

A	Circle; $r = 8\cos\theta$
B	Circle; $r = 4\cos\theta$
C	Parabola; $r = \tan\theta \sec\theta$
D	Hyperbola; $r = \sec^2\theta$

Identify the graph of each rectangular equation. Then write the equation in polar form. Support your answer by graphing the polar form of the equation.

4A. $x^2 + (y - 3)^2 = 9$

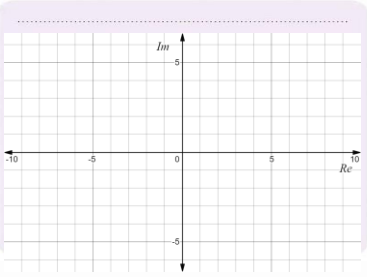
A	Hyperbola; $r^2\cos2\theta = 1$
B	Circle; $r = 6\sin\theta$
C	Parabola; $r = \tan\theta\sec\theta$
D	Hyperbola; $r = \sec^2 \theta$

4B. $x^2 - y^2 = 1$

A	Hyperbola; $r^2\cos2\theta = 1$
B	Circle; $r = 6\sin\theta$
C	Parabola; $r = \tan\theta\sec\theta$
D	Hyperbola; $r = \sec^2 \theta$

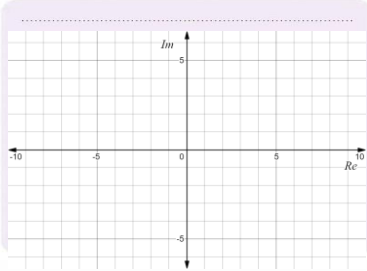
Graph each number in the complex plane, and find its absolute value.

a. $z = 4 + 3i$



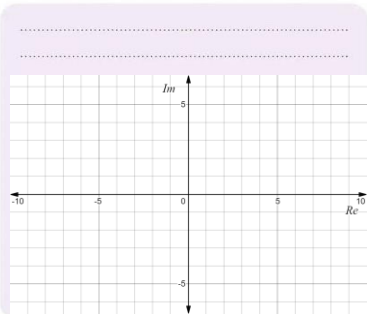
A	$ z =\sqrt{5}$
B	$ z =5$
C	$ z =\sqrt{29}$
D	$ z =25$

b. $z = -2 - i$



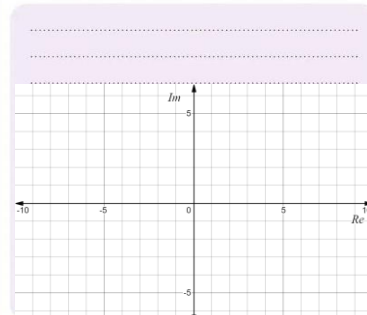
A	$ z =\sqrt{5}$
B	$ z =5$
C	$ z =\sqrt{29}$
D	$ z =25$

1A. $5 + 2i$



A	$ z =\sqrt{5}$
B	$ z =5$
C	$ z =\sqrt{29}$
D	$ z =25$

1B. $-3 + 4i$



A	$ z =\sqrt{5}$
B	$ z =5$
C	$ z =\sqrt{29}$
D	$ z =25$

A	$4\sqrt{3} - 4i$
B	$3\sqrt{3} - 4i$
C	$-3.88 + 14.49i$
D	$3.11 + 11.59i$

A	$4\sqrt{3} - 4i$
B	$3\sqrt{3} - 4i$
C	$-3.88 + 14.49i$
D	$3.11 + 11.59i$

A	$4\sqrt{3} - 4i$
B	$3\sqrt{3} - 4i$
C	$-3.88 + 14.49i$
D	$3.11 + 11.59i$

Determine whether each sequence is arithmetic.

a. 5, -6, -17, -28, ...

A	Yes, with a common difference of $d = 16$
B	No, because there is no common difference
C	Yes, with a common difference of $d = -11$
D	Yes, with a common difference of $d = 9$

b. -4, 12, 28, 42, ...

A	Yes, with a common difference of $d = 16$
B	No, because there is no common difference
C	Yes, with a common difference of $d = -11$
D	Yes, with a common difference of $d = 9$

1B. -6, 3, 12, 21, ...

A	Yes, with a common difference of $d = 16$
B	No, because there is no common difference
C	Yes, with a common difference of $d = -11$
D	Yes, with a common difference of $d = 9$

1A. 7, 12, 16, 20, ...

A	Yes, with a common difference of $d = 16$
B	No, because there is no common difference
C	Yes, with a common difference of $d = -11$
D	Yes, with a common difference of $d = 9$

Find the next three terms of each geometric sequence. Then graph the sequence.

14. 8, 12, 18, 27, ...

A	128,256,512
B	40.5, 60.75, 91.1
C	0.4, 0.08, 0.016
D	$\frac{1}{9}, -\frac{1}{27}, \frac{1}{81}$

15. 8, 16, 32, 64, ...

A	128,256,512
B	40.5, 60.75, 91.1
C	0.4, 0.08, 0.016
D	$\frac{1}{9}, -\frac{1}{27}, \frac{1}{81}$

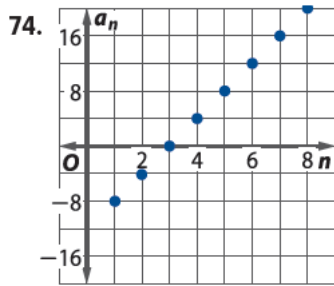
16. 250, 50, 10, 2, ...

A	128,256,512
B	40.5, 60.75, 91.1
C	0.4, 0.08, 0.016
D	$\frac{1}{9}, -\frac{1}{27}, \frac{1}{81}$

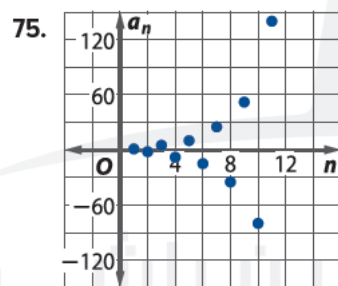
17. 9, -3, 1, $-\frac{1}{3}$, ...

A	$-\frac{1}{9}, -\frac{1}{27}, -\frac{1}{81}$
B	40.5, 60.75, 91.1
C	0.4, 0.08, 0.016
D	$\frac{1}{9}, -\frac{1}{27}, \frac{1}{81}$

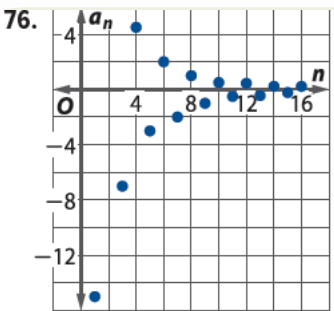
Determine whether each sequence is *convergent* or *divergent*.



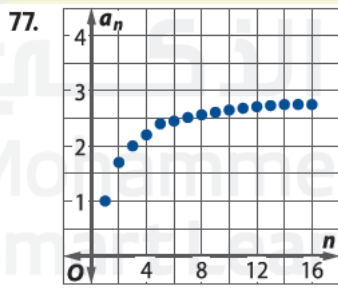
A	Convergent;approaches16
B	Divergent
C	Convergent;approaches0
D	Convergent;approaches3



A	Convergent;approaches16
B	Divergent
C	Convergent;approaches0
D	Convergent;approaches3



A	Convergent;approaches16
B	Divergent
C	Convergent;approaches0
D	Convergent;approaches3



A	Convergent;approaches16
B	Divergent
C	Convergent;approaches0
D	Convergent;approaches3

KeyConcept Sigma Notation

For any sequence $a_1, a_2, a_3, a_4, \dots$, the sum of the first k terms is denoted

$$\sum_{n=1}^k a_n = a_1 + a_2 + a_3 + \dots + a_k,$$

where n is the index of summation, k is the upper bound of summation, and 1 is the lower bound of summation.

16	Use sigma notation to represent and calculate sums of series.	Key Concept • Sigma Notation	552
		(57-62)	554

Write each series in sigma notation. The lower bound is given.

57. $-2 - 1 + 0 + 1 + 2 + 3 + 4 + 5; n = 1$

	A	$\sum_{n=1}^8 (n - 2)$	C	$\sum_{n=1}^8 (n + 1)$
	B	$\sum_{n=1}^8 (n - 3)$	D	$\sum_{n=1}^8 (n - 1)$

58. $\frac{1}{20} + \frac{1}{25} + \frac{1}{30} + \frac{1}{35} + \frac{1}{40} + \frac{1}{45}; n = 4$

	A	$\sum_{n=4}^9 \left(\frac{1}{5n}\right)$	C	$\sum_{n=1}^8 (n + 1)$
	B	$\sum_{n=4}^9 \left(\frac{1}{n}\right)$	D	$\sum_{n=4}^9 \left(\frac{1}{n + 1}\right)$

59. $8 + 27 + 64 + \cdots + 1000; n = 2$

	A	$\sum_{n=4}^9 \left(\frac{1}{5n}\right)$	C	$\sum_{n=2}^{10} (3^n)$
	B	$\sum_{n=2}^{10} (n^2)$	D	$\sum_{n=2}^{10} (n^3)$

60. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{128}; n = 1$

	A	$\sum_{n=1}^{128} \left(\frac{1}{2}\right)^n$	C	$\sum_{n=2}^{10} (3^n)$
	B	$\sum_{n=1}^7 \left(\frac{1}{2}\right)^n$	D	$\sum_{n=3}^9 (-2)^n$

61. $-8 + 16 - 32 + 64 - 128 + 256 - 512; n = 3$

	A	$\sum_{n=1}^{128} \left(\frac{1}{2}\right)^n$	C	$\sum_{n=2}^{10} (3^n)$
	B	$\sum_{n=1}^7 \left(\frac{1}{2}\right)^n$	D	$\sum_{n=3}^9 (-2)^n$

62. $8\left(-\frac{1}{3}\right) + 8\left(\frac{1}{9}\right) + 8\left(-\frac{1}{27}\right) + \cdots + 8\left(-\frac{1}{243}\right); n = 1$

	A	$\sum_{n=1}^5 8\left(\frac{1}{3}\right)^n$	C	$\sum_{n=2}^{10} (3^n)$
	B	$\sum_{n=1}^5 8\left(-\frac{1}{3}\right)^n$	D	$\sum_{n=1}^5 8\left(\frac{1}{3}\right)^{n-1}$

KeyConcept n th Term of an Arithmetic Sequence

The n th term a_n of an arithmetic sequence in which the first term is a_1 and the common difference is d is given by the following formula, where n is any natural number.

$$a_n = a_1 + (n - 1)d$$

Find the sum of each arithmetic series.

57. $\sum_{k=1}^{16} (4k - 2)$

58. $\sum_{k=4}^{13} (4k + 1)$

59. $\sum_{k=5}^{16} (2k + 6)$

60. $\sum_{k=0}^{12} (-3k + 2)$

A	480
B	512
C	544
D	1024

A	350
B	385
C	412
D	700

A	270
B	312
C	324
D	348

A	-240
B	-208
C	-195
D	208

Find three geometric means between 2 and 1250.

A	8 , 32 , 128
B	10 , 50 , 250
C	12 , 60 , 300
D	20 , 100 , 500

3. Find four geometric means between 0.5 and 512.

A	2 , 8 , 32 , 128
B	1 , 4 , 18 , 54
C	4 , 16 , 64 , 256
D	2 , 6 , 18 , 54

Find a_1 in a geometric series for which $S_n = 13,116$, $n = 7$, and $r = 3$.

A	$a_1 = 4$
B	$a_1 = 6$
C	$a_1 = 12$
D	$a_1 = 18$

6. Find a_1 in a geometric series for which $S_n = -26,240$, $n = 8$, and $r = -3$.

A	$a_1 = -16$
B	$a_1 = 8$
C	$a_1 = 16$
D	$a_1 = 32$

Find $\sum_{k=1}^\infty 18\left(\frac{4}{5}\right)^{k-1}$.

A	18
B	72
C	48
D	Not exist

3. Find $\sum_{k=1}^\infty 12\left(\frac{3}{4}\right)^{k-1}$.

A	18
B	72
C	48
D	Not exist

Write a recursive formula for each sequence.

a. 2, 10, 18, 26, 34, ...

A	$a_1 = 2 ; a_n = 2a_{n-1} + 8$
B	$a_1 = 2 ; a_n = a_{n-1} + 8$
C	$a_1 = 2 ; a_n = 2a_{n-1} - 8$
D	$a_1 = 2 ; a_n = a_{n-1} + 10$

2A. 8, 20, 50, 125, 312.5, ...

A	$a_1 = 8 ; a_n = 2a_{n-1}$
B	$a_1 = 8 ; a_n = 2.5a_{n-1}$
C	$a_1 = 8 ; a_n = 4a_{n-1}$
D	$a_1 = 8 ; a_n = a_{n-1} + 16$

2B. 8, 17, 26, 35, 44, ...

A	$a_1 = 8 ; a_n = 2a_{n-1}$
B	$a_1 = 8 ; a_n = 2a_{n-1} + 9$
C	$a_1 = 8 ; a_n = a_{n-1}$
D	$a_1 = 8 ; a_n = a_{n-1}$

2C. $a_3 = 16$ and $r = 4$

A	$a_1 = 1 ; a_n = 2a_{n-1}$
B	$a_1 = 1 ; a_n = 4a_{n-1}$
C	$a_1 = 1 ; a_n = a_{n-1} + 4$
D	$a_1 = 1 ; a_n = 2a_{n-1}+4$

A	6, -10, 38
B	-10, 38, -106
C	14, 74, 324
D	2, 14, 74

- | | |
|----------|----------------------|
| A | 6, -10, 38 |
| B | -10, 38, -106 |
| C | 14, 74, 324 |
| D | 2, 14, 74 |

A	$165y^8z^3$
B	$330y^7z^4$
C	$462y^6z^5$
D	$330y^4z^7$

- | | |
|---|-------------|
| A | $210c^5d^5$ |
| B | $210c^6d^4$ |
| C | $252c^6d^4$ |
| D | $252c^5d^5$ |

Find a counterexample to disprove each statement.

6. $3^n + 1$ is divisible by 4.

A	$n = 1$
B	$n = 2$
C	$n = 3$
D	$n = 5$

7. $2^n + 3^n$ is divisible by 4.

A	$n = 1$
B	$n = 2$
C	$n = 3$
D	$n = 5$

21. $1 + 2 + 3 + \dots + n = n^2$

A	$n = 1$
B	$n = 2$
C	$n = 3$
D	$n = 5$

22. $1 + 8 + 27 + \dots + n^3 = (2n + 2)^2$

A	$n = 1$
B	$n = 2$
C	$n = 3$
D	$n = 5$

23. $n^2 - n + 15$ is prime.

A	$n = 1$
B	$n = 2$
C	$n = 3$
D	$n = 5$

24. $n^2 + n + 23$ is prime.

A	$n = 1$
B	$n = 2$
C	$n = 3$
D	$n = 5$